

Project 1

Predicting Income Using the UCI Adult Income

Mai Nguyen, Isaac Guttormson

The Question

Can census information predict income above \$50K?

- Based on someone's age, education, occupation, and relationship status be used to predict their income?
- Hypothetical user: Targeted advertising to potential customers
 - Decision: Who should we be advertising to?
 - How this model helps: Predicting whether a customer has enough money to buy a product based on their individual income

The Dataset

UCI Adult Income Dataset from the UC Irvine Machine Learning Repository:

- 32561 rows
- 15 Variables
 - age, workclass, final weight, education, education years, marital status, occupation, relationship, race, sex, capital gain, capital loss, hours per week, native country, income
- Target Variable: Income

Models - Setup

- All features were used initially
- Bias and Leakage Tests
 - Sex and Race
 - Capital Gain and Capital Loss
- Issues with the dataset
 - Workclass, Occupation, and Native Country all had missing values
 - Missing values were initially denoted by '?', they were replaced with null values instead of dropping the rows
- Train / Test Split:
 - 26049 Train Rows, 6512 Test Rows
 - 80/20 split

Models - Implementation

Logistic Regression:

- Simple implementation and analysis
- Low computational cost

Random Forest:

- Can handle non-linear relationships better, capture more complicated pattern
- More resistant to overfitting and easier to fine tune

Results

model	n	accuracy	precision	recall	f1	roc_auc	average_precision
Majority class	16281	0.7638	0.0000	0.0000	0.0000	0.5000	0.2362
Logistic regression	16281	0.8487	0.7204	0.5876	0.6473	0.9025	0.7552
Random forest	16281	0.8649	0.7839	0.5913	0.6741	0.9157	0.7979

model	split	n	accuracy	precision	recall	f1
Majority class	train	26049	0.7592	0.0000	0.0000	0.0000
Majority class	validation	6512	0.7592	0.0000	0.0000	0.0000
Logistic regression	train	26049	0.8489	0.7297	0.5916	0.6534
Logistic regression	validation	6512	0.8606	0.7531	0.6263	0.6838
Random forest	train	26049	0.8791	0.8281	0.6282	0.7145
Random forest	validation	6512	0.8745	0.8106	0.6250	0.7058

What the Model Serves

Possible Application:

- Marketing
 - Predicting a potential customers income can be useful insight into whether or not the potential customer has the money to spend on a product
 - False Positive: (predict >\$50K but actually earn \$50K or less)
 - The model falsely predicts a customer makes over \$50k a year, this leads to a waste of money since the customer is less likely to purchase the product
 - False Negative: (actually earn >\$50K but model predicts they are not)
 - The model falsely predicts that a potential customer makes under \$50k a year, this leads to a lack of marketing towards the customer and can lose a potential sale

What the Model Serves

Optimizing the Model for this Scenario:

- Optimizing Precision:
 - This would lead to more false negatives, leading to possible losses in sales by not advertising to the optimal amount of possible customers. This would be better if a company had a small marketing budget and had to be very precise in scouting and marketing to potential customers
- Optimizing Recall:
 - This leads to more false positives, which would be wasting money on marketing on unlikely customers. This is better for companies with larger marketing budgets, where publicity is more important than searching for potential customers

Generally, optimizing precision is more efficient as it leads to less money spent on advertising

Performance at Different Thresholds

Logistic Regression at Different Thresholds

Threshold	split	n	accuracy	precision	recall	f1
0.3	train	26049	0.8314	0.6221	0.7654	0.6942
0.3	validation	6512	0.8401	0.6378	0.7811	0.7121
0.5	train	26049	0.8489	0.7297	0.5916	0.6534
0.5	validation	6512	0.8606	0.7531	0.6263	0.6838
0.7	train	26049	0.8392	0.8477	0.4011	0.5562
0.7	validation	6512	0.8542	0.8517	0.4549	0.5981

Random Forest at Different Thresholds

Threshold	split	n	accuracy	precision	recall	f1
0.3	train	26049	0.8714	0.6912	0.8451	0.7518
0.3	validation	6512	0.8421	0.6484	0.7861	0.7006
0.5	train	26049	0.8791	0.8281	0.6282	0.7145
0.5	validation	6512	0.8745	0.8106	0.6250	0.7058
0.7	train	26049	0.8542	0.8977	0.4066	0.5504
0.7	validation	6512	0.8398	0.8615	0.3981	0.5342

A lower threshold increases recall while precision decreases, a higher threshold increases precision at the cost of recall, in our example

Choose the right threshold based on the company's goal

What the Model is Doing

Top Five Features:

1. Marital Status: 0.120
2. Relationship: 0.113
3. Education Years: 0.092
4. Capital Gain: 0.092
5. Occupation: 0.061

Permutation importance measures how much validation drops when a feature is shuffled. Larger drops mean the model relies more on that feature.

What the Model is Doing

- Specific Person -

Wrong Prediction:

In this instance, our Forest model gives this person a total contribution of +0.16265 based on individual variables

This is added to the 0.24 of the dataset above \$50k to get ~0.40

Since our threshold is 0.5 and $0.4 < 0.5$, this person is predicted to earn below \$50k when actually they earn above \$50k

Feature	Value	Contribution
age	28	-0.10842
marital_status	Married-civ-spouse	+0.09789
relationship	Husband	+0.07300
workclass	Local-gov	+0.06953
occupation	Protective-serv	+0.05181
capital_gain	0	-0.02459
hours_per_week	40	-0.01703
sex	Male	+0.01429
education_num	12	+0.01170
capital_loss	0	-0.00899
native_country	United-States	+0.00207
race	White	+0.00139
Total		+0.16265

What the Model is Doing - Specific Person

Wrong Prediction:

Here we see that the biggest negative contributors to the wrong prediction are:

Age of 28 with -0.10842

Capital Gain of 0 with -0.02459

Hours Per Week at 40 with -0.01703

Feature	Value	Contribution
age	28	-0.10842
marital_status	Married-civ-spouse	+0.09789
relationship	Husband	+0.07300
workclass	Local-gov	+0.06953
occupation	Protective-serv	+0.05181
capital_gain	0	-0.02459
hours_per_week	40	-0.01703
sex	Male	+0.01429
education_num	12	+0.01170
capital_loss	0	-0.00899
native_country	United-States	+0.00207
race	White	+0.00139
Total		+0.16265

Judgment - Sex and Race

Precision: Fairly consistent at about 78% with an outlier for the 'other' race at 90%

Recall: Much more varied across sex and race, specifically with American Indian Eskimo

attribute	group	n	positives	base_rate	selection_rate	precision	recall	fpr	recall_ci_low	recall_ci_high
sex	Female	5421	590	0.1088	0.0701	0.7921	0.5102	0.0164	0.4699	0.5503
sex	Male	10860	3256	0.2998	0.2321	0.7826	0.6060	0.0721	0.5891	0.6226
race	Amer-Indian-Eskimo	159	19	0.1195	0.0503	0.7500	0.3158	0.0143	0.1536	0.5399
race	Asian-Pac-Islander	480	133	0.2771	0.2208	0.7642	0.6090	0.0720	0.5241	0.6878
race	Black	1561	179	0.1147	0.0737	0.7826	0.5028	0.0181	0.4303	0.5752
race	Other	135	25	0.1852	0.0815	0.9091	0.4000	0.0091	0.2340	0.5926
race	White	13946	3490	0.2503	0.1908	0.7843	0.5980	0.0549	0.5816	0.6141

Judgment - What Was Done

Retraining without Sex and Race increases the gap in the Recall between male and female by about 0.002

It also increased Recall for American Indian Eskimo (+0.1) and Asian Pacific Islander (+0.03) and Other (+0.08)

attribute	group	n	positives	selection_rate	recall	fpr	recall_ci_low	recall_ci_high
sex	Female	5421	590	0.0697	0.5102	0.0159	0.4699	0.5503
sex	Male	10860	3256	0.2340	0.6081	0.0738	0.5912	0.6247
race	Amer-Indian-Eskimo	159	19	0.0692	0.4211	0.0214	0.2314	0.6372
race	Asian-Pac-Islander	480	133	0.2354	0.6316	0.0836	0.5470	0.7088
race	Black	1561	179	0.0769	0.5028	0.0217	0.4303	0.5752
race	Other	135	25	0.0963	0.4800	0.0091	0.3003	0.6650
race	White	13946	3490	0.1909	0.5980	0.0550	0.5816	0.6141

Judgment - Testing Leakage

Testing the differences when removing Sex and Race results in little to no change in any of the metrics

Without Capital Gains and Losses almost all metrics decrease by a noticeable amount

model	split	accuracy	precision	recall	f1
Full features	validation	0.8745	0.8106	0.6250	0.7058
Full features	test	0.8649	0.7839	0.5913	0.6741
Without capital gains and losses	validation	0.8521	0.7494	0.5797	0.6537
Without capital gains and losses	test	0.8385	0.7103	0.5343	0.6099
Without sex and race	validation	0.8750	0.8060	0.6333	0.7093
Without sex and race	test	0.8647	0.7814	0.5931	0.6744

Judgment - Ship It?

Both models are fairly simple and the threshold can be easily changed to fit a more specific scenario, they should be easy to implement and do significantly better than the 76% majority class

The Random Forest was only slightly better than the Regression, it would have been interesting to see a more complex model perform with a significantly higher accuracy

Judgment - Limitations

Both models are relatively simple, they both got around at most 85-87% accuracy and it seemed difficult to break that threshold

The dataset itself is quite outdated, a more current version could have better insight, specifically in bias testing

Conclusion

- Census information can help predict whether someone earns above \$50K.
- Random forest performed better than logistic regression, reaching **86.5% accuracy and 0.674 F1** at the 0.50 threshold.
- However, it identified only **59.1% of actual higher-income people**, so many predictions still need improvement.
- Error rates differed across demographic groups, and removing sex and race did not eliminate the gap.